

1. A quadratic function passes through the coordinates (2, 13), (1, 2), and (-2, 41).

- (a) Construct a system of linear equations using these points and the general formula for a quadratic function. **[3 marks]**

$$\begin{cases} 4a + 2b + c = 13 \\ a + b + c = 2 \\ 4a - 2b + c = 41 \end{cases}$$

- (b) Write the augmented matrix of the system. **[3 marks]**

$$\left[ \begin{array}{ccc|c} 4 & 2 & 1 & 13 \\ 1 & 1 & 1 & 2 \\ 4 & -2 & 1 & 41 \end{array} \right]$$

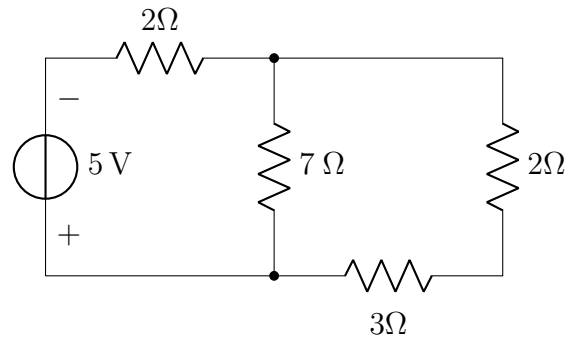
- (c) Use elementary row operations to bring the augmented matrix to Reduced Row Echelon Form. **[6 marks]**

$$\left[ \begin{array}{ccc|c} 1 & 0 & 0 & 6 \\ 0 & 1 & 0 & -7 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

- (d) Calculate the quadratic function. **[3 marks]**

$$f(x) = 6x^2 - 7x + 3$$

2. Consider the electrical circuit below.



- (a) Choose directions for two mesh currents,  $I_1$  and  $I_2$ , one for each loop, and clearly state your convention. **[2 marks]**

$I_1 \rightarrow$  Clockwise current in left Loop  
 $I_2 \rightarrow$  Clockwise current in right Loop

- (b) Write the resulting system of linear equations for the unknown mesh currents  $I_1$  and  $I_2$ . **[3 marks]**

$$I_1 : 9I_1 - 7I_2 = 5$$

$$I_2 : -7I_1 + 12I_2 = 0$$

- (c) Write the augmented matrix of the system. **[3 marks]**

$$\begin{bmatrix} 9 & -7 & 5 \\ -7 & 12 & 0 \end{bmatrix}$$

- (d) Use elementary row operations to bring the augmented matrix to Reduced Row Echelon Form. **[5 marks]**

$$\begin{bmatrix} 1 & 0 & \frac{60}{59} \\ 0 & 1 & \frac{35}{59} \end{bmatrix}$$

- (e) Calculate the currents flowing through this circuit. **[2 marks]**

$$\begin{array}{l} I_1 = 1.017A \\ I_2 = 0.593A \end{array} \quad \text{middle current} = 0.424A$$

3. Consider the two matrices below.

$$A = \begin{bmatrix} 2 & -5 \\ 7 & 0 \\ 3 & 2 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 0 & 6 \\ -3 & 2 & 0 \end{bmatrix}$$

(a) Compute  $AB$ .

[4 marks]

$$AB = \begin{bmatrix} 17 & -10 & 12 \\ 7 & 0 & 42 \\ -3 & 4 & 18 \end{bmatrix}$$

(b) Compute the transpose of  $AB$ .

[2 marks]

$$\begin{bmatrix} 17 & 7 & -3 \\ -10 & 0 & 4 \\ 12 & 42 & 18 \end{bmatrix}$$

(c) Compute the transpose of both  $A$  and  $B$ .

[4 marks]

$$\begin{bmatrix} 2 & 7 & 3 \\ -5 & 0 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3 \\ 0 & 2 \\ 6 & 0 \end{bmatrix}$$

(d) Compute  $A^T B^T$ .

[4 marks]

$$\begin{bmatrix} 20 & 8 \\ 7 & 15 \end{bmatrix}$$

(e) Compute  $B^T A^T$ .

[4 marks]

$$\begin{bmatrix} 17 & 7 & -3 \\ 10 & 0 & 4 \\ 12 & 42 & 18 \end{bmatrix}$$

(f) Compare your answers for this question. What do your results imply about the algebraic properties of a transposition? [2 marks]

the transpose of a matrix product reverses the order of multiplication

4. (a) Calculate the determinant of  $A$ .

[2 marks]

$$A = \begin{bmatrix} 6 & 2 \\ 21 & 7 \end{bmatrix}$$

$$\det(A) = (6 \times 7) - (2 \times 21) = 0$$

- (b) Compute the inverse of  $A$ .

[5 marks]

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- (c) Explain, is  $A$  an elementary matrix? Why?

[3 marks]

NO, every elementary matrix is invertible. Since  $\det(A) = 0$

- (d) Calculate the determinant of  $B$ .

[3 marks]

$$B = \begin{bmatrix} 4 & 2 & 0 \\ 5 & -3 & 2 \\ -1 & 3 & -2 \end{bmatrix}$$

$$\det(B) = 4(-3(-2) - 2(3)) - 2(5(-2) - 2(-1)) + 0 = 16$$

- (e) Compute the inverse of  $B$ . **[8 marks]**

$$\begin{bmatrix} 0 & 2 & 2 \\ 4 & -4 & -4 \\ 6 & -7 & -11 \end{bmatrix}$$

- (f) Compute  $B^{-1}B$ . **[4 marks]**

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

